Construction Chemicals

SUBMITTAL DOCUMENT

O-Swell

Hydrophilic Swelling Waterstop
for Construction Joints



Swelling
1000 – 1300 %

Water Pressure
Up to 6 bar

Easy Install
No welding

Re-swelling
Wet / dry cycles

Project	
Main Contractor	
Consultant	
Submittal No. / Rev.	00
Date	Nov 2026

Construction Chemicals

Contents

1. Method Statement 3

2. Technical Data Sheet 9

3. Test Reports 12

4. Approvals & Certificates 13

METHOD STATEMENT

O-Swell — Hydrophilic Swelling Waterstop for Construction Joints

Project Name		**Document No.**	OMX-MS-OSWELL
Main Contractor		**Revision**	01
Consultant		**Date of Issue**	Nov 2026
Product	O-Swell	**Status**	For Approval

1. Purpose

This method statement defines the procedure, sequence, resources, quality checks and safety measures for the supply and application of O-Swell on site. It ensures that the work is carried out safely, in line with the approved drawings and project specification, and achieves the performance stated in the O MIX Technical Data Sheet.

2. Scope of Work

This method statement covers surface preparation, fixing and protection of O-Swell hydrophilic swelling waterstop in construction joints and around pipe penetrations of concrete structures.

- Inspection and preparation of the joint surface.
- Fixing the swelling strip by nails, profile channel or adhesive.
- Detailing at pipe penetrations and strip ends.
- Inspection before concreting.
- Protection of the strip until the next pour.

3. Definitions

Term	Meaning
Swelling waterstop	Strip that expands on contact with water to seal a joint.
Construction joint	Joint between two pours without designed movement.
Concrete cover	Concrete thickness between the strip and the nearest surface.
Premature swelling	Swelling before the strip is embedded, caused by rain or standing water.

4. Roles & Responsibilities

Role	Responsibility
Project Manager	Overall responsibility for planning, resources, approvals and coordination with the consultant and main contractor.
Site Engineer	Supervises the work daily; confirms substrate readiness, climate conditions, mixing, consumption and sequence; keeps daily records.

Role	Responsibility
QA/QC Inspector	Carries out inspections and tests at each stage, records results, raises inspection requests at hold points and signs the checklist.
HSE Officer	Conducts toolbox talks, checks PPE, ventilation, access and safe storage; stops work if unsafe conditions arise.
Foreman / Applicators	Carry out the work as per this method statement and report any deviation immediately.
Storekeeper	Receives, stores and issues materials in original packaging under the stated storage conditions; controls batch numbers.

5. Materials

Material	Description
O-Swell	Swelling waterstop, 20 × 5 mm or 20 × 10 mm, rolls of 10 m
Fixing materials	Concrete nails, profile channels or compatible adhesive, tie wire

5.1 Key product data

Property	Typical Value
Material	Hydrophilic synthetic rubber (butyl / bentonite)
Free swelling in water	1000 – 1300 %
Swelling in rain water	1100 %
Swelling in concrete pore water	950 %
Hardness	Shore A 35
Tensile strength	> 2 MPa
Elongation	> 400 %
Water pressure resistance	up to 6 bar

6. Tools & Equipment

- Wire brush
- Blower or vacuum
- Hammer
- Knife
- Tape measure
- Thermo-hygrometer and surface thermometer (dew point check)
- Moisture meter
- Measuring buckets and scale
- Masking tape and protective sheeting
- Lighting and extension cables

6.1 Personal protective equipment

- Gloves
- Safety glasses
- Helmet
- Safety shoes

7. Storage & Handling

- Store in original cartons in a dry place, protected from water, rain and humidity, at +5 °C to +35 °C.
- Store on pallets, off the ground, in a dry, covered and ventilated area.
- Keep containers closed until use; use materials on a first-in, first-out basis.
- Store rolls and cartons flat, off the ground, protected from direct sunlight, heat, oil and mechanical damage.
- Handle with care to avoid damage to packaging; do not stack beyond the safe height.

8. Pre-Start Requirements

- Material submittal, this method statement and the inspection plan approved by the consultant.
- Work area handed over clean, free of other trades and with safe access.
- Adequate lighting, power supply and ventilation available.
- Climate conditions checked and recorded before work starts.
- A trial area prepared and approved where required by the project.
- Toolbox talk held with the work team covering this method statement and the safety measures.

9. Surface Preparation

1. Check the joint position and confirm that 60 mm minimum concrete cover is available on both sides.
2. Remove laitance, loose particles, dust and standing water from the joint surface.
3. Level uneven surfaces so the strip is in full contact with the concrete.
4. Obtain QA/QC approval of the prepared joint before fixing.

10. Preparation of Material

1. Check the size and condition of the rolls; reject swollen or wet material.
2. Keep rolls in their cartons until needed.
3. Cut the required lengths with a sharp knife just before installation.

11. Installation

1. Place the strip in the centre of the joint surface.
2. Fix with concrete nails approximately every 300 mm, in a profile channel, or bond with adhesive.
3. Butt-join ends tightly without gaps.

4. At pipe penetrations, wrap one full turn around the pipe and secure with tie wire.
5. Pour the next stage of concrete as soon as possible, ensuring careful compaction around the strip.
6. Record the installed length per joint.

12. Protection

- Protect the strip from rain and standing water until it is embedded in concrete.
- Cover installed strips with plastic sheeting if rain is expected.
- Replace any strip that has swollen before concreting.

13. Quality Control & Inspection

Check	Stage	Acceptance Criteria	By
Joint surface	Before fixing	Clean, sound, even, no standing water	QA/QC
Cover	Before fixing	Min. 60 mm on both sides	Site Engineer
Fixing	After installation	Nails at ~300 mm; full contact	QA/QC
Ends	After installation	Tightly butt-joined	QA/QC
Condition	Before concreting	Dry, not swollen	Site Engineer

13.1 Inspection and test plan

Activity	Inspection / Test	Frequency	Record
Material receipt	Size, condition, dry	Each delivery	Material inspection report
Joint preparation	Clean, sound, even	Each joint	Inspection request (hold point)
Installation	Position, fixing centres, butt joints	Each joint	Inspection request
Before concreting	Strip dry, not swollen	Each pour	Pour record

14. Defects & Remedies

Defect	Likely Cause	Remedy
Premature swelling	Exposure to rain or standing water	Remove and replace the strip
Strip displaced	Insufficient fixing	Re-fix at 300 mm centres
Gaps at ends	Strip not butt-joined	Re-cut and butt-join tightly
Insufficient cover	Strip too close to the surface	Relocate to give 60 mm cover

15. Health, Safety & Environment

Hazard	Control Measures
Cuts from knives and tools	Cut-resistant gloves; sharp tools kept in holders

Hazard	Control Measures
Work in excavations / at height	Shoring, approved access, harness where required
Manual handling of rolls	Correct lifting technique; two-person lift for heavy rolls
Slips and trips	Housekeeping; barricades and signs

- Wear gloves when handling and cutting.
- Use safety glasses when nailing.
- Follow site safety rules for work in excavations.
- Eyes: rinse with clean water.
- Skin: wash with soap and water.

15.1 Waste & environment

- Collect empty containers and residues in designated bins.
- Do not discharge material or washings into drains or soil.
- Allow small quantities of leftover mixed material to cure before disposal; dispose of according to local regulations.

16. Cleaning

- Clean tools with a dry cloth; no cleaning agent required.

17. Limitations

- Not for expansion or moving joints.
- Minimum concrete cover 60 mm.
- Do not install in standing water or in rain.
- Any deviation from this method statement must be approved by the Site Engineer and the consultant.

Appendix A — Inspection Checklist

Inspection Item	Result	Remarks
Joint position and cover verified	☐ Yes ☐ No	
Joint surface cleaned	☐ Yes ☐ No	
Strip in centre of joint	☐ Yes ☐ No	
Fixed at ~300 mm centres	☐ Yes ☐ No	
Ends tightly butt-joined	☐ Yes ☐ No	
Pipe penetrations wrapped	☐ Yes ☐ No	
Strip dry before concreting	☐ Yes ☐ No	
Next pour placed promptly	☐ Yes ☐ No	

Inspected by	Signature	Date

Construction Chemicals

Technical Data Sheet

O-Swell — Submittal Document

Construction Chemicals

Product Datasheet
Edition: 01 | Date: Nov 2026
Waterproofing

O-Swell

Hydrophilic Swelling Waterstop
for Construction Joints



Product Description

O-Swell is a German-technology hydrophilic swelling waterstop strip based on synthetic rubber (butyl / bentonite). On contact with water it expands by 1000 – 1300 % and seals construction joints, pipe penetrations and gaps in concrete. It re-swells after drying, needs no welding and is suitable for fresh, saline and waste water.

Uses

- Horizontal and vertical construction joints.
- Pipe penetrations through concrete walls and slabs.
- Joints between precast elements.
- Basements, tanks, tunnels and foundations.
- Structures exposed to sea water or sewage.

Advantages

- Swells 1000 – 1300 % on contact with water.
- Seals against water pressure up to 6 bar.
- No welding needed; quick installation.
- Re-swells after repeated wet/dry cycles.
- Suitable for saline and waste water.

Application

1. **Surface Preparation**
   Clean, sound and even concrete surface at the joint; remove laitance and loose particles.
2. **Fixing**
   Fix with concrete nails every ~300 mm, in a profile channel, or bond with adhesive; butt-join ends without gaps.
3. **Concreting**
   Keep minimum 60 mm concrete cover; pour the next stage before the strip is exposed to water.

Consumption

1 m per 1 m of joint
plus butt-joint overlap
For pipe penetrations, one full turn around the pipe.

Packaging

Rolls of 10 m; 5 rolls per carton. Sizes 20 × 5 mm and 20 × 10 mm.

Storage & Shelf Life

Store in original cartons in a dry place, protected from water, rain and humidity, at +5 °C to +35 °C.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
Material	Hydrophilic synthetic rubber (butyl / bentonite)	—
Free swelling in water	1000 – 1300 %	—
Swelling in rain water	1100 %	—
Swelling in concrete pore water	950 %	—
Hardness	Shore A 35	—
Tensile strength	> 2 MPa	—
Elongation	> 400 %	—
Water pressure resistance	up to 6 bar	—
Service temperature	-20 °C to +75 °C	—
Minimum concrete cover	60 mm	—

Consumption

Size (width × thickness)	Roll length	Pack
20 × 5 mm	10 m	5 rolls / carton
20 × 10 mm	10 m	5 rolls / carton

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- For construction joints only; not for expansion or moving joints.
- Minimum concrete cover over the strip: 60 mm.
- Substrate clean, sound and even.
- Protect the strip from water and rain before concreting to prevent premature swelling.
- Pour the next concrete stage as soon as possible after installation.

Application Notes

- Fix with concrete nails approximately every 300 mm, in a profile channel, or with a compatible adhesive.
- Butt-join strip ends tightly; do not leave gaps.
- On pipe penetrations, wrap fully around the pipe and secure with tie wire.
- Install in the centre of the joint where possible.
- Do not install in standing water.

Health & Safety

- Wear gloves when handling and cutting.
- Use safety glasses when nailing.
- Follow site safety rules for work in excavations.

First Aid & Cleaning

- Eyes: rinse with clean water.
- Skin: wash with soap and water.
- Cuts: clean and dress the wound.
- Clean tools with a dry cloth; no cleaning agent required.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.

Construction Chemicals

Test Reports

O-Swell — Submittal Document

To be attached: laboratory test reports supporting the values in the Technical Data Sheet.

Construction Chemicals

Approvals & Certificates

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To be attached: approvals and certificates, where available.